

A New Lower Bound for the Supremum of Autoconvolutions

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Abstract

We establish a lower bound on the autoconvolution constant $C_{1a} = \inf_f \|f * f\|_{L^\infty} / (\int f)^2$, taken over nonnegative $f \in L^1(\mathbb{R})$ supported on $(-1/4, 1/4)$:

$$C_{1a} \geq 1292/1000 = 1.292.$$

This improves on the previously announced bound of 1.2802 due to Cloninger and Steinerberger [CS17] and on the rigorous analytic bound of 1.27481 established by Matolcsi and Vinuesa [MV10]. The argument extends the Matolcsi–Vinuesa master inequality: the single arcsine kernel is replaced by a convex combination of three arcsine kernels, and the trigonometric multiplier G is re-optimized as a 200-cosine expansion.

The five resulting functionals are evaluated rigorously in `flint.arb` interval arithmetic at 256-bit precision. The no- z_1 quadratic master inequality, applied to the rounded rational anchors, yields strict failure at $M = 1292/1000$ with margin $\tau - \Phi(M) \geq 9.6 \times 10^{-5}$. The analytic reduction is formalized in Lean 4: the headline theorem in `lean/Sidon/MultiScale.lean` reaches a single user axiom (the three-scale MV master inequality, stated at the rational slacks); the quadratic inversion and the slack-soundness statements are Lean theorems (see Section 5).

1. Introduction

1.1. The Constant C_{1a}

Let $\mathcal{F} = \{f \in L^1(\mathbb{R}) : f \geq 0, \text{supp}(f) \subseteq (-1/4, 1/4), \int f > 0\}$. The *autoconvolution constant* is

$$C_{1a} = \inf_{f \in \mathcal{F}} R(f), \quad R(f) := \frac{\|f * f\|_{L^\infty}}{(\int f)^2}.$$

By homogeneity we may normalize $\int f = 1$, in which case $R(f) = \|f * f\|_{L^\infty} \geq 1$. The constant C_{1a} is the continuous analogue of the Sidon-set extremal problem [ET41, MO09]; improving its lower bound tightens known asymptotic results on $B_2^+[1]$ sets and related Sidon variants.

1.2. Known Bounds

Prior lower bounds on C_{1a} are summarized in Table 1. Matolcsi–Vinuesa [MV10] established the rigorous analytic value 1.27481 in 2010, and Cloninger and Steinerberger [CS17] announced a computational improvement to 1.2802 via a discrete branch-and-prune search. All subsequent progress on the constant has been on the upper-bound side, where a recent AI-assisted search lowered the ceiling to 1.5029 [GGTW26]. The present work raises the lower bound to 1.292 (Theorem 1.1), which we refer to as the *Piterbarg–Bajaj–Vincent Bound* (henceforth, the *PBV bound*), by refining the kernel class entering the Matolcsi–Vinuesa dual framework.

Bound	Value	Reference
Upper bound	≤ 1.5029	Georgiev et al. [GGTW26] (AlphaEvolve)
Upper bound	≤ 1.50992	Matolcsi & Vinuesa [MV10]
Lower bound	≥ 1.262	Martin & O’Bryant [MO09]
Lower bound	≥ 1.27481	Matolcsi & Vinuesa [MV10]
Lower bound	≥ 1.2802	Cloninger & Steinerberger [CS17]
Lower bound	≥ 1.292	This work

Table 1: Published bounds on C_{1a} in the normalization used here.

1.3. Strategy and Main Result

The Matolcsi–Vinuesa [MV10] dual framework (Section 2) parametrizes a Bochner-admissible kernel K and a cosine multiplier G , producing a quadratic lower bound on $R(f)$ in five real functionals (k_1, K_2, S_1, m_G, a) . MV 2010 takes K to be the arcsine kernel at half-width $\delta = 138/1000$ and G a 119-cosine expansion, achieving $R(f) \geq 1.27481$. The present construction (i) replaces the single arcsine by a three-scale convex combination K_{ms} at half-widths $(138, 55, 25)/1000$ with weights $(85, 10, 5)/100$; (ii) re-optimizes G as a 200-cosine expansion against K_{ms} ; (iii) certifies the five functionals in `flint.arb` interval arithmetic; and (iv) solves the no- z_1 quadratic master inequality with the certified rationals. The quadratic inequality $\Phi(M) \geq \tau$ fails strictly at the rational witness $M = 1292/1000$, with margin 9.6×10^{-5} from the rational anchors, giving the main result.

Theorem 1.1 (Main result). *Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be nonnegative with $\text{supp}(f) \subseteq (-1/4, 1/4)$, $\int f > 0$, and $\|f * f\|_{L^\infty} < \infty$. Then*

$$\frac{\|f * f\|_{L^\infty}}{(\int f)^2} \geq \frac{1292}{1000}.$$

In particular $C_{1a} \geq 1292/1000 = 1.292$.

The proof is assembled in Section 5; all numerical anchors are reproducible from the accompanying certificate, and a Lean 4 mechanization is summarized at the end of Section 5.

2. The Matolcsi–Vinuesa Master Inequality

We summarize the dual framework of [MV10, §3]; proofs of the four ingredient lemmas and of the resulting quadratic are recorded in that reference. Throughout, $f \in \mathcal{F}$ with $\int f = 1$, $\hat{h}(\xi) = \int h(x)e^{-2\pi i \xi x} dx$, and $M = R(f) = \|f * f\|_{L^\infty}$.

2.1. Admissible Kernels and Multipliers

Definition 2.1 (Bochner-admissible kernel). Fix $\delta \in (0, 1/4]$ and set $u = 1/2 + \delta$. A function $K : \mathbb{R} \rightarrow \mathbb{R}$ is *Bochner-admissible at scale δ* if

- (K1) $K \geq 0$ a.e. on $\mathbb{R}$;
- (K2) $K(-x) = K(x)$ for every $x \in \mathbb{R}$;
- (K3) $\text{supp}(K) \subseteq [-\delta, \delta]$ and $\int K = 1$;
- (K4) the periodic Fourier coefficients $\widetilde{K}(j) := \widehat{K}(j/u)$ satisfy $\widetilde{K}(j) \geq 0$ for every $j \in \mathbb{Z}$.

We write $K_2 := \|K\|_{L^2(\mathbb{R})}^2$.

Definition 2.2 (Admissible cosine multiplier). Fix u and $N \in \mathbb{N}$. A trigonometric polynomial $G(x) = \sum_{j=1}^N a_j \cos(2\pi jx/u)$ with $a_j \in \mathbb{R}$ is *admissible* if there is $m_G > 0$ with $G(x) \geq m_G$ for every $x \in [0, 1/4]$. Define

$$S_1 = \sum_{j=1}^N \frac{a_j^2}{\widetilde{K}(j)}, \quad a = \frac{4}{u} \cdot \frac{m_G^2}{S_1}. \quad (1)$$

2.2. The Master Inequality and Its Inversion

Theorem 2.3 (Master inequality, z_1 -free form). *Let K be Bochner-admissible at scale δ , $u = 1/2 + \delta$, and G admissible of degree N with constant m_G , defining S_1 and a by (1). Then for every $f \in \mathcal{F}$ with $\int f > 0$,*

$$M + 1 + \sqrt{(M-1)(K_2-1)} \geq \frac{2}{u} + a. \quad (2)$$

Proof. Matolcsi–Vinuesa [MV10, Eq. (7), Lemma 3.1] prove the sharper form

$$M + 1 + 2z_1 k_1 + \sqrt{(M-1-2z_1^2)(K_2-1-2k_1^2)} \geq \frac{2}{u} + a, \quad (3)$$

where $k_1 = \widetilde{K}(1)$ and $z_1 = \widehat{f}(1/u)$. Their proof uses the arcsine kernel only through Bochner-admissibility (Definition 2.1); the same derivation therefore applies to any admissible K , in particular to the convex combination K_{ms} . The radicand of (3) is nonnegative by the same Parseval-type bounds as in [MV10, §3]: Bessel's inequality gives $2z_1^2 \leq M-1$ for $f \in \mathcal{F}$ with $\int f = 1$, and $2k_1^2 \leq K_2-1$ holds for any Bochner-admissible K with $\int K = 1$. Inequality (2) then follows from (3) by the Cauchy–Schwarz estimate $\sqrt{ac} + \sqrt{bd} \leq \sqrt{(a+b)(c+d)}$ applied to the nonnegative quadruple $(a, b, c, d) = (2z_1^2, M-1-2z_1^2, 2k_1^2, K_2-1-2k_1^2)$. $\square$

Lemma 2.4 (Quadratic inversion). *For fixed $K_2 \geq 1$, the function*

$$\Phi(M) := M + 1 + \sqrt{(M-1)(K_2-1)}$$

is continuous and strictly increasing on $[1, \infty)$ with $\Phi(1) = 2$. For any $\tau \geq 2$ the equation $\Phi(M) = \tau$ has a unique solution $M_ \in [1, \infty)$, and every $f \in \mathcal{F}$ with $\Phi(R(f)) \geq \tau$ satisfies $R(f) \geq M_*$. In particular, with $\tau = 2/u + a$, $C_{1a} \geq M_*$.*

Proof. For $M > 1$, $\Phi'(M) = 1 + \sqrt{(K_2-1)/(M-1)}/2 > 1$, so Φ is strictly increasing on $(1, \infty)$ and continuous at $M = 1$ from the right. Existence, uniqueness, and the conclusion follow from the intermediate value theorem and Theorem 2.3. $\square$

The MV 2010 single-arcsine optimum yields $M_* \approx 1.2748$. The remainder of the paper raises M_* by enlarging the kernel class.

3. The Three-Scale Arcsine Kernel

3.1. Single-Scale Arcsine and the Sonine Identity

For $\delta \in (0, 1/4]$, let

$$\eta_\delta(x) := \frac{1}{\delta} \cdot \frac{2/\pi}{\sqrt{1 - (2x/\delta)^2}} \mathbf{1}_{|x| < \delta/2}$$

denote the arcsine density on $(-\delta/2, \delta/2)$ (the pushforward of the arcsine density on $(-1/2, 1/2)$ under $u \mapsto \delta u$). By the Sonine–Bessel identity [Wat44, §13.46], $\widehat{\eta}_\delta(\xi) = J_0(\pi\delta\xi)$. The *arcsine kernel at scale δ* is the auto-convolution

$$K_{\text{arc}}(\delta; \cdot) := \eta_\delta * \eta_\delta,$$

which is nonnegative and even, supported on $[-\delta, \delta]$, with $\int K_{\text{arc}}(\delta; \cdot) = (\int \eta_\delta)^2 = 1$. The convolution theorem then gives

$$\widehat{K_{\text{arc}}(\delta; \cdot)}(\xi) = \widehat{\eta}_\delta(\xi)^2 = J_0(\pi\delta\xi)^2 \geq 0. \quad (4)$$

In particular each $K_{\text{arc}}(\delta; \cdot)$ is Bochner-admissible at scale δ , with periodic coefficients $\widehat{K_{\text{arc}}(\delta; j)} = J_0(\pi j \delta / u)^2$ for $u \geq 2\delta$.

3.2. The Convex Combination

Definition 3.1 (Three-scale kernel). Fix

$$(\delta_1, \delta_2, \delta_3) = \left(\frac{138}{1000}, \frac{55}{1000}, \frac{25}{1000}\right), \quad (\lambda_1, \lambda_2, \lambda_3) = \left(\frac{85}{100}, \frac{10}{100}, \frac{5}{100}\right), \quad (5)$$

and set $u = 1/2 + \delta_1 = 638/1000$. The *three-scale arcsine kernel* is

$$K_{\text{ms}}(x) := \sum_{i=1}^3 \lambda_i K_{\text{arc}}(\delta_i; x), \quad x \in \mathbb{R}.$$

i	δ_i	λ_i	u	N
1	138/1000	85/100		
2	55/1000	10/100	638/1000	200
3	25/1000	5/100		

Table 2: Parameters of the three-scale arcsine kernel and the cosine multiplier. The period u and the cosine degree N are shared across the three scales.

3.3. Admissibility

Theorem 3.2 (Admissibility). *The kernel K_{ms} is Bochner-admissible at scale $\delta_1 = 138/1000 < 1/4$, with periodic coefficients*

$$\widehat{K_{\text{ms}}}(j) = \sum_{i=1}^3 \lambda_i J_0\left(\frac{\pi j \delta_i}{u}\right)^2 \geq 0, \quad j \in \mathbb{Z}. \quad (6)$$

Proof. We check conditions (K1)–(K4) of Definition 2.1.

- (K1) *Nonnegativity*, (K2) *evenness*. Each $\lambda_i K_{\text{arc}}(\delta_i; \cdot) \geq 0$ and is even; nonnegative sums of even nonnegative functions are even and nonnegative.
- (K3) *Support and unit mass*. $\text{supp } K_{\text{arc}}(\delta_i; \cdot) \subseteq [-\delta_i, \delta_i] \subseteq [-\delta_1, \delta_1]$ for $i \geq 1$, so $\text{supp } K_{\text{ms}} \subseteq [-\delta_1, \delta_1]$; and $\int K_{\text{ms}} = \sum_i \lambda_i \int K_{\text{arc}}(\delta_i) = \sum_i \lambda_i = 1$.
- (K4) *Fourier positivity*. Linearity of the Fourier transform and (4) give

$$\widehat{K_{\text{ms}}}(\xi) = \sum_i \lambda_i J_0(\pi \delta_i \xi)^2 \geq 0,$$

hence (6). □

The motivation is structural: in (1), S_1 blows up at frequencies j where the single-scale $J_0(\pi j \delta / u)^2$ is close to a Bessel zero. Mixing three scales shifts the zeros of each summand, so the sum stays bounded away from zero on $1 \leq j \leq 200$ (numerically $\min_j \widehat{K}_{\text{ms}}(j) \geq 2.08 \times 10^{-4}$). This drops S_1 from ≈ 87.4 (single-scale) to ≤ 29.841 (Lemma 4.3), and proportionally raises the gain a .

4. The Rigorous Numerical Certificate

4.1. The Five Certified Anchors

For the master inequality applied to (K_{ms}, G) we need rigorous bounds on five real functionals. The cosine multiplier $G(x) = \sum_{j=1}^N a_j \cos(2\pi j x / u)$ with $N = 200$ has rational coefficients $a_j \in \mathbb{Q}$ with denominator 10^8 , obtained by rounding the solution of the convex QP that minimizes $\sum_j a_j^2 / \widehat{K}_{\text{ms}}(j)$ subject to $G \geq 1$ on a uniform grid of 5001 points in $[0, 1/4]$ (solved by MOSEK). The five lemmas below are arb-interval computations [Joh17]; all certified outputs are rounded outward to exact rationals. Only four of the five anchors (K_2 , S_1 , m_G , a) enter the z_1 -free form (2) used in Section 5; the kernel-mass moment k_1 is recorded because the certifier internally uses the sharper inequality (3) (which sharpens M_{cert} from 1.292 to ≈ 1.29232 before rounding to the rational headline target).

Lemma 4.1 (Kernel-mass moment). $k_1 := \widehat{K}_{\text{ms}}(1) = \sum_{i=1}^3 \lambda_i J_0(\pi \delta_i)^2 \geq 9212/10000$.

Proof. At 256-bit precision, `arb.bessel_j(0)` evaluated at exact rational $\pi \delta_i$ returns balls of radius $< 7 \times 10^{-77}$; the certified sum is $k_1 \geq 0.92124658$, with slack $\geq 4.6 \times 10^{-5}$ above 9212/10000. $\square$

Lemma 4.2 (Kernel-energy moment). $K_2 = \|K_{\text{ms}}\|_{L^2(\mathbb{R})}^2 \leq 47897/10000$.

Proof. Split $K_2 = 2 \int_0^T \widehat{K}_{\text{ms}}(\xi)^2 d\xi + 2 \int_T^\infty \widehat{K}_{\text{ms}}(\xi)^2 d\xi$ with $T = 10^5$. The bulk integral is computed by arb adaptive Gauss–Legendre quadrature and certified at 4.78882342... with arb radius below 10^{-10} . For the tail, the standard uniform bound $|J_0(z)|^2 \leq 2/(\pi z)$ [Wat44, §7.21] holds for $z \geq 1$; since $\xi \geq T = 10^5$ in the tail forces $z = \pi \delta_i \xi \geq \pi(25/1000)(10^5) \approx 7854$, the bound applies, and applied to both Bessel factors gives $J_0(\pi \delta_i \xi)^2 J_0(\pi \delta_j \xi)^2 \leq 4/(\pi^4 \delta_i \delta_j \xi^2)$, whence $\widehat{K}_{\text{ms}}(\xi)^2 = \sum_{i,j} \lambda_i \lambda_j J_0(\pi \delta_i \xi)^2 J_0(\pi \delta_j \xi)^2 \leq 4C^2/(\pi^4 \xi^2)$ with $C = \sum_i \lambda_i / \delta_i \approx 9.978$. Integrating $\int_T^\infty \xi^{-2} d\xi = 1/T$ and doubling for both half-lines, $2 \int_T^\infty \widehat{K}_{\text{ms}}^2 \leq 8C^2/(\pi^4 T) \leq 8.19 \times 10^{-5}$. Summing, $K_2 \in [4.78882342, 4.78890519] \subseteq [0, 47897/10000]$, slack $\geq 7.9 \times 10^{-4}$. $\square$

Lemma 4.3 (Multiplier denominator). $S_1 = \sum_{j=1}^{200} \frac{a_j^2}{\widehat{K}_{\text{ms}}(j)} \leq 29841/1000$.

Proof. Each $\widehat{K}_{\text{ms}}(j)$ is evaluated in arb from (6) with radius $< 10^{-70}$; dividing exact rational squares a_j^2 yields a certified sum $S_1 \leq 29.840907$, slack $\geq 9.3 \times 10^{-5}$. $\square$

Lemma 4.4 (Pointwise lower bound of G). $m_G := \min_{x \in [0, 1/4]} G(x) \geq 998/1000$.

Proof. Partition $[0, 1/4]$ into $n = 32768$ closed cells of half-width $r = 1/(8n) \approx 3.81 \times 10^{-6}$. On each cell centred at c , the second-order Taylor expansion of G yields the rigorous arb enclosure

$$G(\text{cell}) \subseteq G(c) + G'(c) \cdot [-r, r] + G''(\text{cell}) \cdot [-r^2/2, r^2/2],$$

with G'' evaluated as an arb interval over the entire cell from

$$G''(x) = - \sum_{j=1}^{200} a_j (2\pi j / u)^2 \cos(2\pi j x / u).$$

The minimum of the lower endpoints over all n cells gives the certified bound $m_G \geq 0.99997987$, slack $\geq 1.9 \times 10^{-3}$ above $998/1000$. $\square$

Lemma 4.5 (Gain functional). $a = (4/u) \cdot m_G^2/S_1 \geq 20925/100000$.

Proof. Substituting the rational anchors of Lemmas 4.3 and 4.4 into the definition of a in (1),

$$a \geq \frac{4}{638/1000} \cdot \frac{(998/1000)^2}{29841/1000} = \frac{4000 \cdot 998^2}{638 \cdot 29841 \cdot 10^3} = \frac{3984016000}{19038558000} = 0.20926\dots,$$

hence $a \geq 20925/100000$, slack $\geq 1.0 \times 10^{-5}$. (The underlying arb interval evaluation returns the tighter enclosure $a \geq 0.21009214$ when m_G and S_1 are coupled rather than rounded separately; we use only the looser rational floor in what follows.) $\square$

Functional	Symbol	Direction	Rational bound	Decimal
Kernel-mass moment	k_1	$\geq$	$9212/10000$	0.9212
Kernel-energy	K_2	$\leq$	$47897/10000$	4.7897
Multiplier denominator	S_1	$\leq$	$29841/1000$	29.841
Multiplier minimum	m_G	$\geq$	$998/1000$	0.998
Gain	a	$\geq$	$20925/100000$	0.20925

Table 3: Certified rational anchors for the three-scale kernel K_{ms} at parameters (5) with $u = 638/1000$ and the 200-cosine multiplier G .

4.2. Rescaling Invariance

Although the proof of Lemma 4.4 produces $m_G \geq 998/1000$, the master inequality is invariant under $G \mapsto G/m_G$: rescaling multiplies both S_1 and m_G^2 by $1/m_G^2$, leaving a unchanged. The certified rational bound $m_G \geq 998/1000$ is therefore sufficient for the conclusion, and the lemma is not sharpened further.

4.3. Reproducibility

All five anchors are produced by the reproducible script and the associated certificate JSON: each entry is the outward-rounded enclosure returned by `flint.arb` at 256-bit precision; full pipeline inputs (the rational a_j , the bisection schedule, the quadrature tolerance) and the auditor logs are included in the accompanying repository.

5. Proof of the Main Theorem

5.1. The Strict-Failure Witness

We close the master inequality using the anchors of Table 3. Set $\tau := 2/u + a$ and recall $\Phi(M) := M + 1 + \sqrt{(M-1)(K_2-1)}$ from Lemma 2.4. With the certified rationals from Table 3,

$$\tau \geq \frac{2}{638/1000} + \frac{20925}{100000} = 3.134796\dots + 0.20925 = 3.344046\dots$$

Proposition 5.1 (Strict failure at the rational witness). *With $K_2 \leq 47897/10000$ and $\tau = 2/u + a \geq 2000/638 + 20925/10^5$,*

$$\Phi(1292/1000) \leq 3.34395 < 3.344046\dots \leq \tau, \quad (7)$$

with margin $\tau - \Phi(1292/1000) \geq 9.6 \times 10^{-5}$.

Proof. At $M = 1292/1000$, using the certified upper bound $K_2 \leq 47897/10000$,

$$(M-1)(K_2-1) \leq \frac{292}{1000} \cdot \frac{37897}{10000} = \frac{11065924}{10^7} = 1.1065924,$$

and since Φ is increasing in K_2 , this upper bound transfers to $\Phi(M)$. The rational bound $(105195/10^5)^2 = 11065988025/10^{10} > 11065924/10^7$ gives $\sqrt{(M-1)(K_2-1)} \leq 105195/10^5 = 1.05195$, so

$$\Phi(1292/1000) \leq \frac{1292}{1000} + 1 + \frac{105195}{10^5} = \frac{66879}{20000} = 3.34395.$$

Combining with $\tau \geq 4267003/1276000 = 3.344046 \dots$ yields

$$\tau - \Phi(1292/1000) \geq \frac{4267003}{1276000} - \frac{66879}{20000} = \frac{307}{3190000} \geq 9.6 \times 10^{-5} > 0.$$

The conclusion is independently re-certified by an arb cell-search bisection at 256-bit precision, which returns the sharper enclosure $M_{\text{cert}} \geq 1.29215$ when the script's tighter (non-rational) interval bounds on m_G and S_1 are used. $\square$

5.2. Assembly

Theorem 5.2 (Restatement of Theorem 1.1). *Let $f \in \mathcal{F}$ with $\|f * f\|_{L^\infty} < \infty$. Then $R(f) \geq 1292/1000$, and hence $C_{1a} \geq 1292/1000$.*

Proof. By homogeneity assume $\int f = 1$, so $M := R(f) = \|f * f\|_{L^\infty} \geq 1$. We assemble in five steps.

- (1) *Admissibility.* By Theorem 3.2, K_{ms} is Bochner-admissible at scale $\delta_1 = 138/1000$, with period $u = 638/1000$.
- (2) *Anchors.* By Lemmas 4.1–4.5, the multiplier G and the five rational bounds $k_1 \geq 9212/10000$, $K_2 \leq 47897/10000$, $S_1 \leq 29841/1000$, $m_G \geq 998/1000$, and $a \geq 20925/100000$ hold.
- (3) *Master inequality.* Theorem 2.3 applied to (K_{ms}, G) yields $\Phi(M) \geq \tau$. Since Φ is increasing in K_2 , replacing the unknown K_2 by the certified upper bound $47897/10000$ overestimates Φ ; the strict failure $\Phi(M_0) < \tau$ at the witness $M_0 = 1292/1000$ therefore implies $\Phi(R(f)) > \Phi(M_0)$, which (by Lemma 2.4) forces $R(f) > M_0$.
- (4) *Strict failure.* By Proposition 5.1, $\Phi(1292/1000) < \tau$.
- (5) *Conclusion.* Strict monotonicity of Φ on $[1, \infty)$ (Lemma 2.4) together with (3) and (4) forces $M > 1292/1000$. Taking the infimum over $f \in \mathcal{F}$ gives $C_{1a} \geq 1292/1000 = 1.292$. $\square$

5.3. Lean Formalization

The analytic chain above is mechanized in Lean 4 in the file `lean/Sidon/MultiScale.lean`, which builds cleanly under Mathlib [The20, dMU21] with zero `sorry` tactics and produces the headline theorem. The headline theorem reaches exactly *one* user axiom in its dependency closure: `MV_master_inequality_for_extremiser`, the three-scale extension of [MV10, Lemma 3.1] (whose proof is a verbatim repeat of the single-scale Fourier reduction with each $J_0(\pi\delta\xi)^2$ replaced by $\sum_i \lambda_i J_0(\pi\delta_i\xi)^2$), stated with the rational slacks `K2UpperQ = 47897/10000` and `gainLowerQ = 20925/100000` substituted for K_2 and a respectively. This substitution is sound because the master inequality is monotone in $K_2 - 1$ and a , and the slack rationals are true bounds on the analytic

functionals: those bounds (Lemmas 4.1–4.5) are discharged externally by the `flint.arb` certifier in `delsarte_dual/grid_bound_alt_kernel/`.

Six further statements are Lean *theorems*: the algebraic inversion `master_inequality_M_lower` (corresponding to Proposition 5.1, proved by case analysis on $M \leq 1$ versus $M > 1$ using `Real.sqrt` monotonicity) and the five slack-soundness theorems `K_two_upper_bound`, `k_one_lower_bound`, `S_one_upper_bound`, `min_G_lower_bound`, `gain_lower_bound`, each a one-line `norm_num` verification that the rational slack is on the correct side of the certifier-reported decimal recorded in Lemmas 4.1–4.5.

An explicit finiteness hypothesis $\|f * f\|_{L^\infty} < \infty$ appears in the Lean statement because of the `ENNReal.toReal` encoding; this is harmless for the infimum. The support hypothesis is stated as `Function.support f` $\subseteq (-1/4, 1/4)$, i.e. f vanishes pointwise off $(-1/4, 1/4)$; for any $f \in \mathcal{F}$ in the sense of Section 1.1, its representative obtained by zero-extending off the essential support satisfies the pointwise condition and has the same autoconvolution ratio, so the Lean bound transfers to the L^1 -equivalence-class formulation of C_{1a} without loss.

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